

EXPERIMENT 2: STUDY OF RTL NOR GATE & TTL NAND GATE

OBJECTIVE:

- To study and understand the internal circuit configuration and operation of the RTL (Resistor-Transistor Logic) NOR gate.
- To analyze the TTL (Transistor-Transistor Logic) NAND gate circuit, especially focusing on the totem-pole output stage.

INTRODUCTION:

Logic families are categorized by their circuit designs, voltage levels, and switching characteristics. RTL uses resistors and bipolar junction transistors (BJTs) as switching devices. It was an early logic family that later evolved into diode-transistor logic (DTL) and transistor-transistor logic (TTL).

TTL is a widely used digital logic family where BJTs perform both the logic and amplification functions. TTL gates often use a totem-pole output stage, which provides faster switching and better drive capabilities.

This experiment focuses on:

- Understanding the RTL NOR gate operation.
- Investigating the TTL NAND gate operation and the role of the totem-pole output stage.

LIST OF EQUIPMENT:

- DC Power Supply (+5V)
- Breadboard
- Resistors (5 pieces)
- Diode (1 piece)
- Transistors (BJTs) (5 pieces)
- Multimeter
- Connecting wires

CIRCUIT DIAGRAM FOR RTL NOR GATE:

TRUTH TABLE:

CIRCUIT DIAGRAM FOR TTL NAND GATE:

TRUTH TABLE:

JUSTIFICATION / EXPLANATION OF THE EXPERIMENTAL DATA:

1. RTL NOR Gate

Concept:

The RTL NOR gate uses resistors as input networks connected to the base of a transistor. When all inputs are LOW, the transistor is OFF, and the output is pulled HIGH through a resistor. If any input is HIGH, it forward-biases the base-emitter junction, turning the transistor ON and pulling the output LOW.

Experimental Result:

- Output is HIGH only when all inputs are LOW.
- Output goes LOW when any input is HIGH.

This behavior matches the NOR gate truth table and shows how the transistor acts as a switch controlled by the input voltages through the resistors.

2. TTL NAND Gate with Totem-Pole Output

Concept:

The TTL NAND gate employs multiple BJTs for logic operations. The totem-pole output stage uses two transistors stacked vertically to provide a low-impedance path to either the positive voltage or ground, improving output drive and switching speed.

The diode D1 protects the output transistor and helps in the switching process by preventing unwanted feedback or saturation.

Experimental Result:

- Output is LOW only when all inputs are HIGH.
- Output is HIGH in all other input combinations.
The measured outputs match the NAND logic truth table, confirming the TTL NAND gate operation with totem-pole output.

ANSWERS TO REPORT QUESTIONS:

1. Analyze all the circuits with the help of the experimental data:

The RTL NOR gate operates by using resistors and a single transistor. It outputs HIGH only when all inputs are LOW, demonstrating NOR logic. The transistor acts as a switch controlling the output voltage.

The TTL NAND gate uses multiple transistors, including a totem-pole output stage for better drive and switching characteristics. It outputs LOW only when all inputs are HIGH, showing NAND logic.

2. What are the features and advantages of TTL gates?

- Fast switching speed: Due to transistor amplification and the totem-pole output stage.
 - Good noise immunity: TTL circuits can tolerate certain noise levels without erroneous switching.
 - High output drive capability: Totem-pole output can source and sink current effectively, driving multiple inputs.
 - Standardized voltage levels: Typically +5V logic levels, ensuring compatibility.
 - Low power dissipation: Compared to earlier logic families like RTL and DTL.
 - Ease of cascading: TTL outputs can easily drive multiple TTL inputs.
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3. What is the function of diode D1 in Fig. 2(b)?

The diode D1 in the TTL NAND gate with totem-pole output prevents the lower transistor from saturating deeply, which can slow switching speed due to stored charge. It also blocks current flowing backward from the output transistor to the previous stage during switching, improving performance and protecting the transistor.

4. Describe the configuration of the totem-pole stage.

The totem-pole output stage consists of two BJTs connected in series vertically at the output: one transistor connects to the positive supply (pull-up transistor) and the other connects to ground (pull-down transistor). Only one transistor conducts at a time, enabling the output node to be driven strongly HIGH or LOW. This configuration provides low output impedance, faster switching, and better drive capability compared to open-collector outputs.

DISCUSSION:

In this experiment, we successfully studied the RTL NOR gate and TTL NAND gate circuits. The RTL NOR gate demonstrated how resistors and a transistor can implement a NOR function, relying on transistor switching and input resistor networks. The TTL NAND gate showed how multiple transistors combined with a totem-pole output stage enhance performance, offering faster switching and stronger output drive.

The role of diode D1 in the TTL circuit is crucial in preventing transistor saturation and protecting the circuit. Understanding the totem-pole configuration explains why TTL logic gates are faster and more robust than RTL circuits.

This experiment deepened the understanding of digital logic families and how discrete components form the foundation of digital systems used in electronics today.